Teaching Statement

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Teaching Philosophy

My goal as a mathematics educator is to help students develop the ability to think mathematically, approach unfamiliar problems with confidence, and understand how abstract ideas connect to practical applications. I believe that learning mathematics involves more than mastering computational techniques or memorizing formulas. Students should understand why a mathematical method works, recognize when it is appropriate, and develop the reasoning skills needed to apply it in new situations. My teaching philosophy is built on three closely connected principles: conceptual understanding, structured problem solving, and meaningful connections between mathematical theory and applications.

I organize my teaching around a progression from intuition to mathematical formulation, followed by careful computation and interpretation. For example, when introducing derivatives in calculus, I begin with the ideas of changing quantities, average rates of change, and slopes of curves before developing the formal definition of a derivative. Once students understand the underlying concept, differentiation rules become meaningful tools rather than isolated formulas. Similarly, in statistics, I emphasize the distinction between a population and a sample, the interpretation of variability, and the role of uncertainty before introducing procedures for confidence intervals and hypothesis testing.

An important part of this approach is encouraging students to explain their reasoning. A correct numerical answer does not necessarily demonstrate a complete understanding of the mathematical problem. I therefore emphasize the formulation of a solution, the justification of each step, and the interpretation of the result. My objective is to help students progress from following worked examples to independently selecting methods, evaluating their own arguments, and communicating mathematical ideas clearly.

Teaching Experience and Professional Growth

My teaching experience spans undergraduate mathematics instruction in Bangladesh and graduate-level teaching support in the United States. Before beginning my doctoral studies at the University of Houston, I served as a Lecturer at the Canadian University of Bangladesh from January 2018 to December 2021 and at Northern University of Bangladesh from January to July 2022. Across these appointments, I taught a broad range of undergraduate courses, including Calculus, Probability and Statistics, Numerical Analysis, Engineering Mathematics, Fourier and Laplace Transforms, and Complex Variables.

Teaching courses with enrollments of approximately 70–100 students provided valuable experience in course planning, classroom communication, assessment design, and the presentation of mathematical ideas to students with different academic backgrounds. These experiences taught me that effective mathematics instruction requires both subject knowledge and the ability to communicate the same concept through multiple perspectives. An algebraic derivation may clarify an idea for one student, while a geometric interpretation, numerical example, or physical application may provide the necessary understanding for another. I learned to adjust my explanations without compromising mathematical rigor or academic expectations.

Since August 2022, I have continued developing my teaching practice as a Graduate Teaching Assistant at the University of Houston. I currently serve as a Lead Teaching Assistant for Statistics for the Sciences (MATH 3339), where I help coordinate teaching assistants, organize examination grading, prepare homework solutions, and support students through office hours, review sessions, and tutoring. These responsibilities involve close collaboration with faculty members and other teaching assistants to maintain consistent assessment practices and provide effective academic support.

My experiences in these different educational environments have broadened my understanding of teaching. I have learned that successful instruction requires more than a well-prepared lecture. It also depends on organized course materials, clear expectations, accessible explanations, constructive feedback, and consistent communication. I now approach teaching as a continuous process of planning, observing student difficulties, reflecting on instructional practices, and improving how mathematical ideas are communicated.

Connecting Mathematical Concepts, Computation, and Applications

My academic background in applied mathematics strongly influences my approach to teaching. I believe students develop a deeper appreciation of mathematics when they understand how theoretical ideas are used to investigate real problems. Applications can motivate abstract concepts, provide intuition for mathematical definitions, and demonstrate the value of rigorous analytical reasoning.

For example, differential equations become more concrete when students investigate population growth, heat transfer, diffusion, or biological processes. Probability distributions and statistical inference become more meaningful when connected to variability, measurement, and decision-making under uncertainty. Similarly, linear algebra can be motivated through systems of equations, matrix transformations, data analysis, and numerical approximation. These connections help students recognize that different areas of mathematics are not isolated subjects but components of a broader mathematical framework.

My doctoral research in numerical partial differential equations, reduced-order modeling, and PDE-constrained optimization provides a natural foundation for integrating mathematical theory with computation. I routinely use MATLAB, Python, numerical linear algebra, and numerical methods to study complex mathematical models. In courses where computational activities are appropriate, I would incorporate carefully designed examples that allow students to visualize mathematical behavior, compare analytical and numerical solutions, and investigate how changing parameters influences a model.

For instance, in a numerical analysis course, students could compare an exact solution with approximations obtained using different step sizes and examine the relationship between accuracy, convergence, and computational cost. In a differential equations course, a simple population-growth model could be used to connect an analytical solution with a numerical simulation. Such activities encourage students to move between mathematical formulation, analytical reasoning, computation, and interpretation.

I view computational tools as a complement to mathematical understanding rather than a replacement for it. Students should be able to explain the mathematics behind an algorithm, evaluate whether a computed result is reasonable, and recognize the limitations of numerical approximations. Developing these skills prepares students for advanced mathematical study and for applications in science, engineering, statistics, and data-driven disciplines.

Student Engagement and an Inclusive Learning Environment

Students enter mathematics classrooms with different levels of preparation, educational experiences, and confidence. Some are comfortable participating in discussions, while others hesitate to ask questions even when they encounter substantial difficulties. My responsibility as an educator is to create an environment in which students feel comfortable engaging with challenging ideas while understanding that sustained effort and careful reasoning are essential to mathematical progress.

One approach I emphasize is identifying the source of a student's difficulty before providing an explanation. During office hours and tutoring sessions, I ask students to describe their reasoning and explain what they have attempted. Rather than immediately demonstrating a complete solution, I use guiding questions to help identify where their understanding becomes unclear. A difficulty with a calculus problem, for example, may arise from an earlier gap in algebra, functions, or graphical interpretation rather than from the calculus concept itself.

I also believe that mathematical mistakes can become valuable learning opportunities. When students arrive at an incorrect result, I encourage them to examine their argument and identify the step at which the reasoning breaks down. This process develops self-evaluation skills and helps students recognize the difference between a computational error and a conceptual misunderstanding. It also encourages students to approach difficult problems with persistence rather than viewing an incorrect first attempt as a failure.

Having taught in both Bangladesh and the United States, I understand that students may be accustomed to different classroom expectations, instructional styles, and levels of participation. I aim to communicate expectations clearly, provide logically organized explanations, and make academic support accessible through office hours, review sessions, and individual conversations. My goal is to maintain a welcoming classroom while holding all students to meaningful standards of mathematical reasoning and achievement.

Assessment, Feedback, and Continuous Improvement

I consider assessment an essential part of the learning process rather than simply a mechanism for assigning grades. Well-designed homework, quizzes, and examinations should evaluate whether students understand mathematical concepts, can apply appropriate techniques, and are able to explain their reasoning. They should also provide information that helps students and instructors identify areas requiring additional attention.

My responsibilities as a Lead Teaching Assistant at the University of Houston have reinforced the importance of transparent and consistent assessment. Preparing detailed homework solutions and coordinating examination grading require careful attention to mathematical accuracy, appropriate partial credit, and consistent evaluation across students. These responsibilities have strengthened my appreciation for grading practices that recognize the reasoning behind a solution rather than focusing exclusively on the final answer.

I believe that feedback is most useful when students can understand the nature of an error and determine how to improve. For example, if students repeatedly confuse statistical significance with practical significance, or apply a mathematical formula without verifying its assumptions, simply providing the correct answer may not resolve the underlying misunderstanding. A more effective response is to explain why the reasoning is incomplete, revisit the relevant concept, and provide an opportunity to apply the corrected understanding.

In my future courses, I plan to use a combination of regular practice, conceptual questions, structured problem-solving activities, and timely feedback. I would also use patterns in student performance to identify topics that need additional examples or alternative explanations. In this way, assessment can support a continuous cycle of teaching, learning, feedback, and instructional improvement.

Technology and Responsible Use of Artificial Intelligence

Technology offers important opportunities to improve mathematics instruction, particularly through visualization, numerical experiments, and interactive exploration. Programming tools can help students investigate problems that are difficult to analyze entirely by hand and provide a bridge between mathematical theory and scientific applications. At the same time, students must understand that computational output is meaningful only when it is supported by correct mathematical reasoning.

The increasing availability of artificial-intelligence tools introduces both opportunities and challenges for mathematics education. Such tools can provide alternative explanations, generate practice problems, suggest approaches, and support independent exploration. However, they may also produce incorrect mathematical arguments or plausible-looking solutions that contain conceptual errors.

I believe students should learn to use these technologies critically and responsibly. In particular, students should be encouraged to verify generated results, examine assumptions, check intermediate steps, and explain solutions in their own words. I would establish clear expectations regarding appropriate technology use in assignments and assessments, while emphasizing that mathematical reasoning and academic integrity remain central to the learning process.

My broader objective is to prepare students to work confidently with modern computational tools without becoming dependent on them for fundamental mathematical understanding.

Mentoring and Future Teaching Goals

I view mentorship as a natural extension of classroom teaching. Students benefit not only from assistance with mathematical problems but also from guidance in developing effective study habits, exploring mathematical interests, and preparing for academic or professional opportunities. Through tutoring, office hours, and individual conversations, I seek to help students become more independent learners who can evaluate their progress and approach challenging topics systematically.

My teaching background prepares me to contribute to a broad range of undergraduate courses, including Calculus, Differential Equations, Probability and Statistics, Linear Algebra, Numerical Analysis, Engineering Mathematics, and Complex Variables. My research experience also provides a strong foundation for advanced courses in scientific computing, numerical partial differential equations, optimization, mathematical modeling, and inverse problems.

An important future goal is to develop student-centered computational projects that connect concepts from different mathematical courses. For example, students could investigate numerical solutions of ordinary differential equations, compare optimization algorithms, study matrix decompositions, or estimate parameters in simple mathematical models. These activities would illustrate how calculus, linear algebra, statistics, and numerical analysis work together in contemporary applied mathematics.

I am also interested in mentoring undergraduate research and independent studies, as well as master's-level projects in applied and computational mathematics. My research areas offer accessible starting points for student projects involving numerical simulation, mathematical modeling, data fitting, algorithm comparison, and visualization. With appropriate guidance, students can progress from implementing established mathematical methods to formulating questions and conducting independent investigations.

Ultimately, I want my students to develop the confidence, curiosity, and analytical discipline required to learn mathematics beyond a single course. Whether they pursue advanced study or apply mathematics in another field, I hope they leave my classroom with a stronger ability to formulate problems, reason carefully, communicate clearly, and use mathematical and computational tools responsibly.